

**AMENDMENT NO. 1 APRIL 1996**  
**TO**  
**IS 6092 ( Part 5 ) : 1985 METHODS OF SAMPLING AND**  
**TEST FOR FERTILIZERS**

**Part 5 DETERMINATION OF SECONDARY ELEMENTS AND**  
**MICRONUTRIENTS**

*( First Revision )*

*( Page 18, clause 6.1.3.6 ) — Add the following new matter:*

**‘6.1.4 Method D (Atomic Absorption Spectrometer Method)**

**6.1.4.1 Reagents**

**Standard Boron Solution** — Weigh 44.095 g of sodium tetraborate decahydrate ( $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$ ) — AR grade on clean watch glass and transfer it to one litre volumetric flask through the funnel giving several washings to watch glass and funnel with glass distilled water. Make the volume up to the mark. Stopper the flask and shake the solution well. This is 5000 ppm boron solution hereinafter called standard A. This solution should be stored in clean bottle for further use.

**Preparation of Working Standards** — Pipette the following volume of standard A in 50 ml numbered volumetric flasks and make the volume with water.

| <i>Flask No.</i> | <i>Volume of standard<br/>A taken (ml)</i> | <i>Concentration of boron<br/>after making volume<br/>to 50 ml (ppm)</i> |
|------------------|--------------------------------------------|--------------------------------------------------------------------------|
| 1.               | 0.0                                        | 0.0                                                                      |
| 2.               | 4.0                                        | 400                                                                      |
| 3.               | 6.0                                        | 600                                                                      |
| 4.               | 8.0                                        | 800                                                                      |
| 5.               | 10.0                                       | 1 000                                                                    |
| 6.               | 12.0                                       | 1 200                                                                    |
| 7.               | 14.0                                       | 1 400                                                                    |
| 8.               | 16.0                                       | 1 600                                                                    |

## Amend No. 1 to IS 6092 ( Part 5 ) : 1985

### 6.1.4.2 Procedure

**Preparation of sample solution** — Weigh 2.500 g of the material on a clean watch glass and transfer it quantitatively in a 250-ml volumetric flask through the funnel giving repeated washing with water. Make up the volume, stopper the flask and shake well.

**Flamming the solutions** — Flame the standards and the sample solution on atomic absorption spectrophotometer at a wavelength of 249.6 nm using nitrous oxide-acetylene flame.

**Calculations** : Prepare a standard curve of known concentrations of boron solutions by plotting the absorbance value on Y-axis against their respective boron concentration on X-axis. Calculate the percentage boron in the given sample by multiplying boron concentration value obtained from standard curve by 0.01.

*Example:*

|                               |                         |
|-------------------------------|-------------------------|
| Mass of the fertilizer sample | = 2.50 g                |
| Volume made                   | = 250 ml                |
| Dilution factor               | $\frac{250}{2.5} = 100$ |

Reading of sample from atomic absorption =  $Y$

Corresponding concentration value of boron from standard curve against  $Y$  absorbance —  $X$  ppm

percentage of boron in the sample —  $0.01 X$